# SEC2_APM1110-FA2_GROUP_8-DABU, L; LIBRES, B

GitHub: https://github.com/dabulira20/Prob-ProbDi-FAs/tree/main/FA2

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## 1. Problem (#3)

An experiment consists of rolling a fair die 600 times. Using R, we simulate this experiment to calculate the empirical relative frequency of each faces of the die (1 through 6) and estimate the probability of rolling each face. These simulated estimates are then compared against the theoretical probability of 1/6   0.1667.

## 2. R Code and Output

```r
# set.seed ensures reproducibility of the random sample
set.seed(123)
n_rolls <- 600
faces <- 1:6

rolls <- sample(faces, size = n_rolls, replace = TRUE)  # simulate 600 rolls
roll_counts <- table(rolls)                              # count each face
relative_freq <- roll_counts / n_rolls                    # relative frequency
relative_freq
```

```
## rolls
##         1         2         3         4         5         6
## 0.1766667 0.1700000 0.1600000 0.1533333 0.1666667 0.1733333
```

## 3. Results Table

```r
results_table <- data.frame(
  Face               = as.integer(names(roll_counts)),
  Count              = as.integer(roll_counts),
  Relative_Frequency = round(as.numeric(relative_freq), 4),
  Theoretical_Prob   = round(1/6, 4),
  Difference         = round(as.numeric(relative_freq) - 1/6, 4)
)
# Results are based on a fixed seed (123) for reproducibility.
results_table
```

```
##   Face Count Relative_Frequency Theoretical_Prob Difference
```

```
## 1     1    106              0.1767            0.1667      0.0100
## 2     2    102              0.1700            0.1667      0.0033
## 3     3     96              0.1600            0.1667     -0.0067
## 4     4     92              0.1533            0.1667     -0.0133
## 5     5    100              0.1667            0.1667      0.0000
## 6     6    104              0.1733            0.1667      0.0067
```

Since we assume to have used a fair die, the theoretical probability for each face is 1/6. In comparison, all six relative frequencies land close to 1/6, with differences no larger than about $\pm 0.012$ — the small gap expected from a finite sample of 600 fair rolls.

## 4. Quick Check

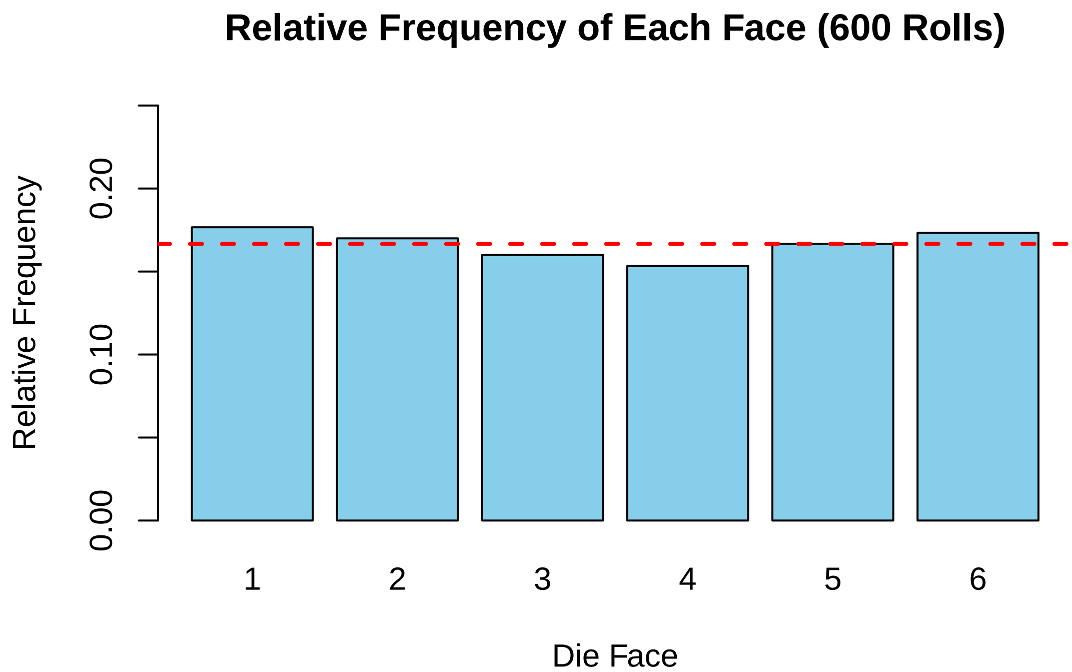
```r
sum(relative_freq)    # should equal 1
```

```
## [1] 1
```

The relative frequencies sum to exactly 1, confirming they form a valid probability distribution and the counts were tabulated correctly.

## 5. Plot

```r
barplot(as.numeric(relative_freq), names.arg = faces, col = "skyblue",
        ylim = c(0, 0.25), xlab = "Die Face", ylab = "Relative Frequency",
        main = paste0("Relative Frequency of Each Face (", n_rolls, " Rolls)"))
abline(h = 1/6, col = "red", lwd = 2, lty = 2)
```




## 6. Conclusion

The simulated relative frequencies for all six faces were close to the theoretical probability of 1/6, confirming that a fair die's outcomes are equally likely and that 600 trials provide a close empirical approximation to the true probability.